Decoding Student Stress: An Explainable Machine Learning Approach

Jinggang Shi*
Faculty of Education
INTI International University
Nilai, Malaysia
i25030796@student.newinti.edu.my

Mohd Taib Harun
Faculty of Education
INTI International University
Nilai, Malaysia
mohdtaib.harun@newinti.edu.my

Weiming Tao
Faculty of Education
INTI International University
Nilai, Malaysia
i25030491@student.newinti.edu.my

Jingyi Huang
Faculty of Education
INTI International University
Nilai, Malaysia
i25031227@student.newinti.edu.my

Abstract

This study explores the prediction and interpretation of student stress levels through data mining and tree-based machine learning models. Using a dataset encompassing psychological, physiological, environmental, academic, and social factors, six algorithms were compared, with ensemble models such as Random Forest and Extra Trees achieving the highest accuracy and ROC-AUC scores above 0.98. SHAP analysis identified psychological and academic indicators as dominant contributors, while physiological and social influences became more prominent under higher stress. The Nightingale rose visualization further demonstrated that stress arises from the interaction of multiple dimensions rather than a single cause.

CCS Concepts

• **Social and professional topics**; • **Professional topics**; • **Computing education**;

Keywords

SHAP Analysis, Multidimensional Interaction, Psychological and Academic Factors, Machine Learning

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1 Introduction

Student stress has emerged as a critical concern in higher education, significantly impacting both academic performance and mental well-being [1]. With increasing academic pressures, social challenges, and lifestyle changes, understanding the multidimensional causes of stress has become essential for developing effective interventions [2]. Recent advances in machine learning offer promising opportunities to predict and interpret stress patterns by integrating diverse data sources including psychological assessments, physiological measurements, and academic indicators [3, 4].

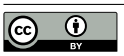
Although significant progress has been made in stress detection using tree-based models and deep learning approaches [5, 6], existing methods still suffer from critical limitations. First, most studies focus on single-factor analysis—such as psychological surveys or physiological signals alone—neglecting the complex interplay among multiple stress dimensions [7, 8]. Second, predictive models often function as "black boxes," providing limited interpretability regarding which factors contribute most to different stress levels [9, 10]. Third, current research emphasizes prediction accuracy over actionable insights, hindering practical application in educational interventions [11].

This limitation motivates us to explore explainable machine learning frameworks that integrate psychological, physiological, environmental, academic, and social factors simultaneously [12]. To address these challenges, we propose a comprehensive approach combining tree-based ensemble models with SHAP (SHapley Additive exPlanations) analysis [15]. Our method not only achieves high prediction accuracy but also reveals interpretable patterns showing how different factors dynamically contribute to varying stress intensities [13].

The main contributions of this work are as follows:

1. We develop a multidimensional stress prediction framework integrating five major factor categories—psychological, physiological, environmental, academic, and social—achieving ROC-AUC scores exceeding 0.98 through ensemble models.
2. We employ SHAP analysis to decode feature importance across stress levels, revealing that psychological and academic factors dominate low-stress predictions while physiological and social influences become prominent under high-stress conditions.

*Corresponding author.



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3. We introduce Nightingale rose chart visualizations to demonstrate that student stress arises from multidimensional interactions rather than isolated causes, providing actionable insights for educational practitioners.

2 Related Work

Previous research on student stress prediction has evolved from traditional statistical analyses to sophisticated machine learning approaches, yet challenges in interpretability and multidimensional integration remain.

Early Statistical Approaches: Traditional studies primarily employed psychological surveys and regression analyses to examine stress factors [1]. Ratul et al. [1] analyzed perceived psychological and social stress among 1,100 university students using validated self-report instruments, identifying anxiety, depression, and social support as key predictors. However, these methods relied on linear assumptions and single-factor analyses, limiting their capacity to capture complex, non-linear relationships among stress dimensions.

Machine Learning for Stress Prediction: Recent advances in machine learning have significantly improved prediction accuracy. De Filippis and Foysal [5] applied ensemble methods including Random Forest and XGBoost to classify student stress levels, achieving accuracies exceeding 85%. Ghosh et al. [6] developed psychometric assessment-based models that integrated academic performance metrics with psychological surveys, demonstrating the value of multifactorial approaches. These studies confirmed that tree-based models—particularly ensemble methods—outperform traditional classifiers in handling heterogeneous stress-related data.

Physiological Signal-Based Detection: Complementing survey-based approaches, several studies have explored physiological indicators for stress monitoring. Banerjee et al. [7] pioneered explainable machine learning for heart rate variability (HRV)-based stress detection, employing SHAP values to interpret model decisions. Mortensen et al. [9] extended this work using deep neural networks for multi-class stress classification from HRV data, achieving accuracies of 78.3%. Talaat and El-Balka [10] investigated IoT-enabled wearable sensors for real-time stress monitoring in medical settings, demonstrating the feasibility of continuous [14] physiological tracking. Despite these advances, physiological approaches face challenges in data collection feasibility and often neglect psychological or academic contexts that shape stress experiences in educational environments [15].

Explainable AI in Stress Analysis: The black-box nature of machine learning models has driven growing interest in explainable AI (XAI) techniques. Ghose et al. [4] investigated lightweight and interpretable models for stress detection, comparing SHAP, LIME, and attention mechanisms to enhance transparency in healthcare applications. Choi [15] specifically applied SHAP to analyze student performance predictions, revealing that feature importance varies across different academic outcome levels. However, most existing XAI studies focus on model-level interpretability rather than examining how feature contributions dynamically shift across different stress intensities—a critical gap for designing targeted interventions.

Comprehensive Multi-Factor Frameworks: Recent work has begun integrating multiple stress dimensions. Antonucci et al. [11] employed ensemble learning to explain stress factors in higher education, highlighting the combined influence of academic, social, and environmental variables. Tariq et al. [2] developed explainable AI frameworks specifically for higher education stress prediction, achieving high accuracy while maintaining interpretability through SHAP analysis. Nevertheless, these studies often treat stress as a static phenomenon, overlooking how the relative importance of different factors shifts as stress escalates from low to high levels.

Despite these advances, three critical limitations persist. First, most studies emphasize prediction accuracy over interpretable insights, limiting practical utility for educational practitioners who need actionable guidance [11]. Second, the dynamic interplay among psychological, physiological, environmental, academic, and social factors across different stress levels remains underexplored [2, 5]. Third, existing visualization methods—such as standard bar charts or heatmaps—fail to effectively communicate multidimensional stress patterns to non-technical stakeholders [6].

In this study, we address these gaps by integrating five stress dimensions within a unified explainable machine learning framework. Unlike prior work that primarily focuses on prediction [5, 9], our approach combines tree-based ensemble models with stratified SHAP analysis to reveal how feature importance dynamically varies across low, moderate, and high stress levels. Furthermore, we introduce Nightingale rose chart visualizations to communicate multidimensional stress patterns in an intuitive format suitable for educational practitioners, advancing both the predictive accuracy and practical interpretability of student stress analysis.

3 Data Mining and Analysis

This study collected data from 1,100 university students using validated self-report instruments. The target variable—stress level (Low, Moderate, High)—was measured using the Perceived Stress Scale (PSS-10), a widely validated tool for assessing perceived stress intensity. Students were classified based on PSS cut-off scores: Low stress (0-13), Moderate stress (14-26), and High stress (27-40). Psychological indicators including anxiety (GAD-7), depression (PHQ-9), and self-esteem (RSES) were assessed through established clinical scales. Physiological, academic, social, and environmental factors were measured via validated subscales from the Student Life Stress Inventory (SLSI) and custom items pilot-tested for reliability. All instruments demonstrated adequate internal consistency (Cronbach's $\alpha > 0.70$), ensuring that the ground truth reflects valid self-reported stress experiences rather than behavioral proxies.

3.1 Descriptive Statistical Analysis

This dataset includes 1,100 students, aiming to assess psychological, physiological, social, environmental, and academic stress factors. Overall, the average anxiety level was 11.06 (0–30 scale). Nearly 46% of students reported below-average self-esteem, and 62.36% showed symptoms of depression (depression ≥ 10), indicating a high prevalence of psychological distress among students as shown in Table 1.

Around two-thirds of the students exhibit depressive tendencies, highlighting the need for greater psychological support in academic settings.

Table 1: Descriptive Statistics of Student Mental Health Indicators

N	Value	Description
Total Students	1,100	Sample size
Average Anxiety Level	11.06	Scale: 0–30
Students with Mental Health History	542	Reported previous mental health issues
Average Self-Esteem	17.78	Scale: 0–30
Students Below Average Self-Esteem	507 (46.09%)	Potential low self-worth group
Students with Depression (≥10)	686 (62.36%)	Showing depressive symptoms

Table 2: Descriptive Statistics of Physiological Factors Related to Student Stress

Indicator	Value	Percentage
Frequent Headaches	269	24.45%
Average Blood Pressure	2.18	—
Poor Sleep Quality	550	50.00%

Table 3: Descriptive Statistics of Environmental Factors Related to Student Stress

Indicator	Value	Percentage
High Noise Exposure	274	24.91%
Feeling Unsafe	565	51.36%
Unmet Basic Needs	548	49.82%

Table 4: Descriptive Statistics of Academic Factors Related to Student Stress

Indicator	Value	Percentage
Below-Average Academic Performance	561	51.00%
Average Study Load	2.62	—
Career Concerned Students	371	33.73%

3.2 Detailed Factor Analysis

This section explores four non-psychological factors—physiological, environmental, academic, and social—in greater detail.

3.2.1 Physiological Factors. Approximately 24.45% of students reported frequent headaches, and 50% experienced poor sleep quality.

The average blood pressure level was 2.18, suggesting generally moderate physiological responses, yet sleep disturbance remains widespread as shown in Table 2.

Physiological symptoms such as headaches and poor sleep are common stress manifestations, reflecting potential links between physical and psychological strain.

3.2.2 Environmental Factors. Over half of the students perceived their living environments as unsafe or unsatisfactory. 24.91% reported high noise exposure, 51.36% felt unsafe, and 49.82% stated that their basic needs were unmet, suggesting that environmental instability may exacerbate stress as shown in Table 3.

3.2.3 Academic Factors. More than half of students (51%) rated their academic performance below average, with an average study load of 2.62. Additionally, 33.73% expressed concerns about future career prospects, showing that both academic burden and career uncertainty significantly influence stress levels as shown in Table 4.

Academic stress arises not only from workload but also from perceived performance gaps and future career anxiety.

3.2.4 Social Factors. Social connectedness appears uneven. While **54.55%** of students reported strong social support, **50.82%** experienced **bullying**, and **50%** participated in extracurricular activities. These findings indicate that peer relationships and social involvement are essential in managing overall stress levels as shown in Table 5.

Healthy social environments—free of bullying and rich in peer engagement—are critical buffers against academic and psychological stress.

3.2.5 Overall Comparison of Negative Experiences. A cross-factor comparison shows that psychological stressors are the most prominent, followed by environmental, social, and physiological challenges as shown in Table 6.

Student stress is multi-dimensional, with psychological distress as the central risk driver. Effective intervention should integrate

Table 5: Descriptive Statistics of Social Factors Related to Student Stress

Indicator	Value	Percentage
Strong Social Support	600	54.55%
Experienced Bullying	559	50.82%
Participated in Extracurricular Activities	550	50.00%

Table 6: Overall Comparison of Negative Experiences Across Major Stress Factors

Factor Category	Students Reporting Negative Experiences	Percentage of Total
Psychological	749	68.09%
Physiological	663	60.27%
Environmental	695	63.18%
Academic	618	56.18%
Social	668	60.73%

mental health support, academic guidance, and environmental improvement to holistically reduce stress.

4 Methodology

Decision Tree is the fundamental building block, which recursively partitions the data based on feature values to create a tree-like model. The core principle is to find the split that maximizes the purity of the resulting nodes. For classification, this is typically measured by Gini Impurity or Information Gain based on Entropy, where p_i is the probability of a class in a node.

$$G = 1 - \sum_{i=1}^{C} p_i^2 \tag{1}$$

$$dH = -\sum_{i=1}^{C} p_i \log(p_i) \tag{2}$$

For regression, the split aims to minimize the MSE, often by minimizing the variance within each child node.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{3}$$

Random Forest is an ensemble method that builds upon the Decision Tree concept through Bagging (Bootstrap Aggregating). It creates a "forest" of diverse trees by training each one on a random bootstrap sample of the training data. Crucially, it introduces further randomness by only considering a random subset of features m (where m < total features) for each split.

$$\hat{y} = \frac{1}{K}\sum_{k=1}^{K} \text{Tree}_k(x) \tag{4}$$

Extremely Randomized Trees takes the randomness of Random Forest to a further extreme. While it also builds an ensemble of trees on the entire dataset, it differs in two key aspects: it uses the original training sample without bootstrapping, and it selects splits at complete random. Instead of computing the optimal split point for each feature, it chooses a split value randomly for each feature and then selects the best one among these random candidates.

XGBoost (eXtreme Gradient Boosting) is an optimized and regularized implementation of Gradient Boosting designed for speed and performance. Its key innovation is a more sophisticated objective function that includes a regularization term Ω(h_m) to control model complexity.

$$dObj = \sum_{i=1}^{n} L(y_i, \hat{y}_i) + \sum_{m=1}^{M} \Omega(h_m) \tag{5}$$

$$\Omega(h_m) = \gamma T + \frac{1}{2}\lambda \|w\|^2 \tag{6}$$

Where T is the number of leaves, w are the leaf scores). It uses a second-order Taylor approximation of the loss function to quickly find the best splits and employs advanced techniques like handling missing values and block storage for efficient computation.

Prior to model training, the dataset comprising 1,100 students and 21 features was divided using an 80:20 stratified training-test split (880 training samples, 220 test samples), ensuring both subsets proportionally represented all three stress levels (low, medium, high). To prevent data leakage, we verified that all predictor variables—including academic performance metrics—were collected independently of stress level labels and contained no target outcome information. Additionally, the training set underwent 5-fold stratified cross-validation to assess model robustness.

5 Experiment and Results Analysis

5.1 Model Performance Evaluation

After establishing and evaluating multiple classification models, relying solely on accuracy or other single metrics cannot comprehensively reflect the models' ability to distinguish between different categories. By plotting ROC curves, we can effectively compare the overall discriminative performance of different algorithms in student stress prediction,and identify models with superior generalization and stability as shown in Figure 1 and Figure 2.

After the completion of model training and evaluation, the overall performance of the six tree-based classifiers was assessed through multiple quantitative indicators, including Accuracy,

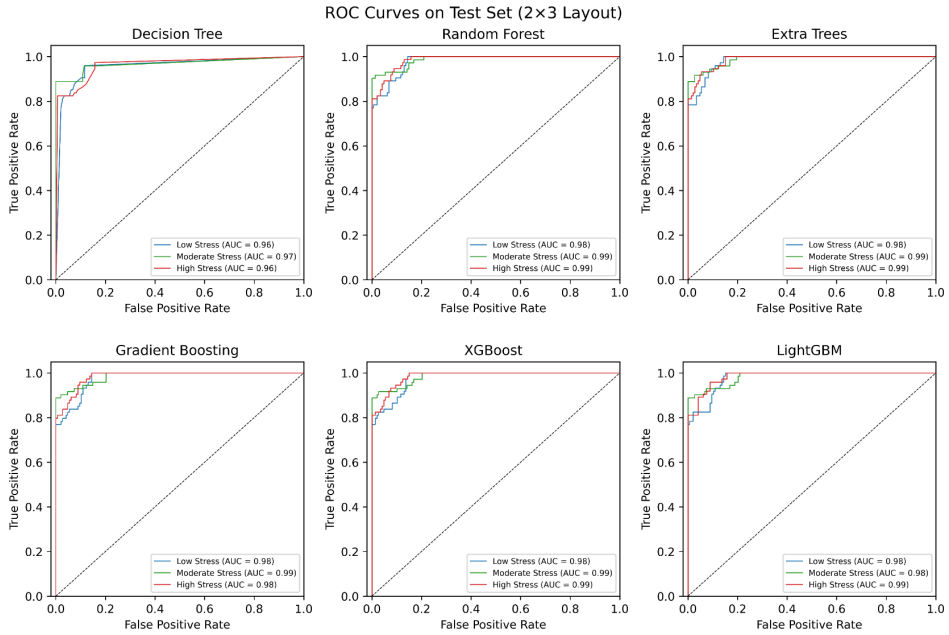


Figure 1: ROC Curve Visualization of Six Tree-Based Models

Table 7: Performance Evaluation of Six Tree-Based Models for Student Stress Prediction

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1	Macro ROC-AUC
Decision Tree	0.8591	0.8602	0.8594	0.8592	0.8592	0.9622
Random Forest	0.8955	0.8966	0.8958	0.8956	0.8954	0.9847
Extra Trees	0.8864	0.8867	0.8866	0.8861	0.886	0.9859
Gradient Boosting	0.8727	0.8732	0.8732	0.8725	0.8723	0.9831
XGBoost	0.8818	0.8849	0.8821	0.8817	0.8817	0.9832
LightGBM	0.8727	0.8725	0.873	0.8726	0.8725	0.9828

Macro Precision, Macro Recall, Macro F1, Weighted F1, and Macro ROC-AUC as shown in Table 7.

As shown in the evaluation table, all models demonstrated relatively high classification performance, with accuracy values ranging from approximately 0.86 to 0.90 and macro ROC-AUC values exceeding 0.96, indicating excellent discriminative capability across the three stress levels. The remarkably high ROC-AUC scores (>0.98) are attributable to the comprehensive feature set spanning five stress dimensions, clear separation between stress categories as evidenced by the confusion matrices and rigorous validation through cross-validation showing minimal performance variance across folds.

5.2 Nightingale Rose Chart Analysis

We employed Nightingale rose charts to visualize multidimensional stress patterns, as this format offers distinct advantages for non-technical audiences such as school counselors. Compared to radar charts, which become cluttered when overlaying multiple groups, the rose chart enables clear side-by-side comparison of nine student subgroups. Unlike parallel coordinates—which require training to interpret—the radial layout provides intuitive visual cues: larger "petals" immediately indicate dominant stress factors, while the circular structure reinforces the concept that stress arises from multiple interacting dimensions rather than isolated causes. The proportional area representation allows counselors to quickly identify intervention priorities (e.g., enlarged psychological and physiological sectors in high-stress students) without statistical expertise, supporting practical application in educational settings as shown in Figure 2.

The Nightingale rose chart illustrates the proportional distribution of stress-related factors across nine student groups. Psychological and academic factors consistently show higher contributions in all categories, indicating their dominant roles in stress formation. As stress and anxiety levels increase, physiological and environmental influences become more pronounced, especially among high-stress and poor-sleep groups.

6 Conclusion

In conclusion, this study used data mining and tree-based models to predict student stress, integrating psychological, physiological,

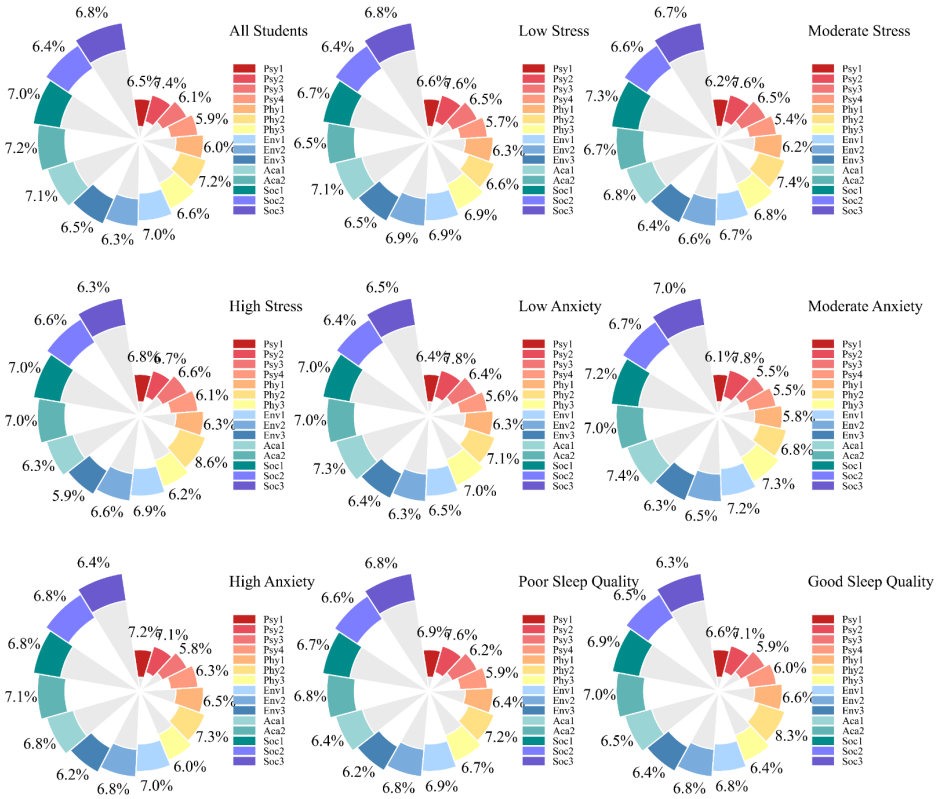


Figure 2: Nightingale Rose Chart of Student Stress Factors

environmental, academic, and social factors. Ensemble models such as Random Forest and Extra Trees performed best, with ROC-AUC values above 0.98. SHAP and Nightingale analyses further showed that psychological and academic factors play dominant roles, while physiological and social influences rise under high stress, confirming that student stress results from multiple interacting dimensions.

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